

ASSESSMENT CompTia N+

Section A - Concept Application

Scenario 1

1. CPU (Central Processing Unit):

- **Socket Compatibility:** The CPU socket on the motherboard must match the CPU model exactly (e.g., Intel LGA 1700 or AMD AM5).

2. RAM (Random Access Memory):

- **Generation Support (DDR4 vs. DDR5):** Your motherboard dictates whether you can use DDR4 or DDR5 RAM—they are physically and electrically incompatible with one another.

3. Storage:

- **Motherboard M.2 Slots and PCIe Lanes:** Verify how many M.2 slots your motherboard has and what PCIe generation they support (e.g., PCIe 4.0 vs.

PCIe 5.0). Using a PCIe 5.0 SSD in a PCIe 4.0 slot will bottleneck its speed to PCIe 4.0 limits.

Scenario 2

- **CPU:** Real-time protection intercepts files the moment you open, modify, or download them to check for threats.
- **RAM:** Antivirus programs keep extensive virus signature databases and active monitoring modules loaded into system memory at all times, reducing the available RAM for other applications.
- **Storage (I/O):** Constant background reading, indexing, and scanning of files generate high disk activity. On traditional hard drives, this causes severe read/write bottlenecks; on SSDs, it creates heavy I/O overhead and temporarily slows down file access speeds.

Scenario 3:

1. Boot into Safe Mode
2. Check Task Manager for Rogue Processes
3. Roll Back or Update Problematic Drivers
4. Run System File Checker (SFC) and DISM

Scenario 4:

1. **Surge Protector:** A surge protector acts as a first line of defense against sudden voltage spikes caused by lightning, downed power lines, or grid switching.
2. **UPS (Uninterruptible Power Supply):** A UPS combines surge protection with a battery backup, offering comprehensive defense against total power loss, brownouts (voltage drops), and sags.

Scenario 5:

DHCP (Dynamic Host Configuration Protocol): Automatically assigns your computer an IP address, subnet mask, default gateway, and crucially, the address of the **DNS server** whenever you connect to the network.

DNS (Domain Name System): Acts as the phonebook of the internet, translating human-readable website names into numeric IP addresses. Your device relies on the DNS server address provided by DHCP to know where to send its lookup requests.

Scenario 6:

1. Temporarily Disable the Firewall:

- Temporarily turn off your firewall (such as Windows Defender Firewall or a third-party security suite) to test if the blocked service or connection starts working.

2. Inspect Firewall Logs:

- Open your firewall's event viewer or logging dashboard to review recently dropped or blocked packets. Look for log entries that match the timestamp of your connection failure and note the blocked IP address, port, or protocol.

3. Test Specific Ports and Services:

- Use diagnostic commands to test specific ports. For example, open PowerShell and run `Test-NetConnection [IP Address] -Port [Port Number]` to see if the traffic is getting through.

4. Create an Explicit Allow Rule:

- Open your firewall configuration settings, create a new rule, and explicitly allow the specific application executable, port number, or remote IP address required for the traffic.

Section B - Practical Application

Task 1:

Workstation Assembly:

1. Install the CPU:

- Place your motherboard on top of its non-conductive cardboard box. Lift the CPU socket retention lever, align the golden triangle on the processor with the triangle on the socket, drop it gently into place without applying force, and lock the lever down.

2. Mount the CPU Cooler:

- Apply a pea-sized drop of thermal paste to the center of the CPU.

3. Prepare the Chassis and Mount the PSU:

- Remove both side panels of your PC case. Slide the Power Supply Unit (PSU) into the bottom rear compartment and secure it with screws from the back of the case.

4. Install the Motherboard:

- Lower the prepared motherboard into the chassis, carefully aligning its rear ports with

the I/O shield opening on the back of the case.

5. Connect Power, Data, and Front Panel Wires:

- Route the main 24-pin ATX power cable and the 8-pin CPU power cable from the PSU through the case cutouts and plug them firmly into their respective motherboard headers.

6. Install the Graphics Card (GPU):

- Remove the necessary PCIe slot covers on the back of the case. Align your graphics card with the top PCIe x16 slot, push it down firmly until the plastic retention clip clicks into place, and secure the metal bracket to the case with a screw.

7. First Boot and POST Check:

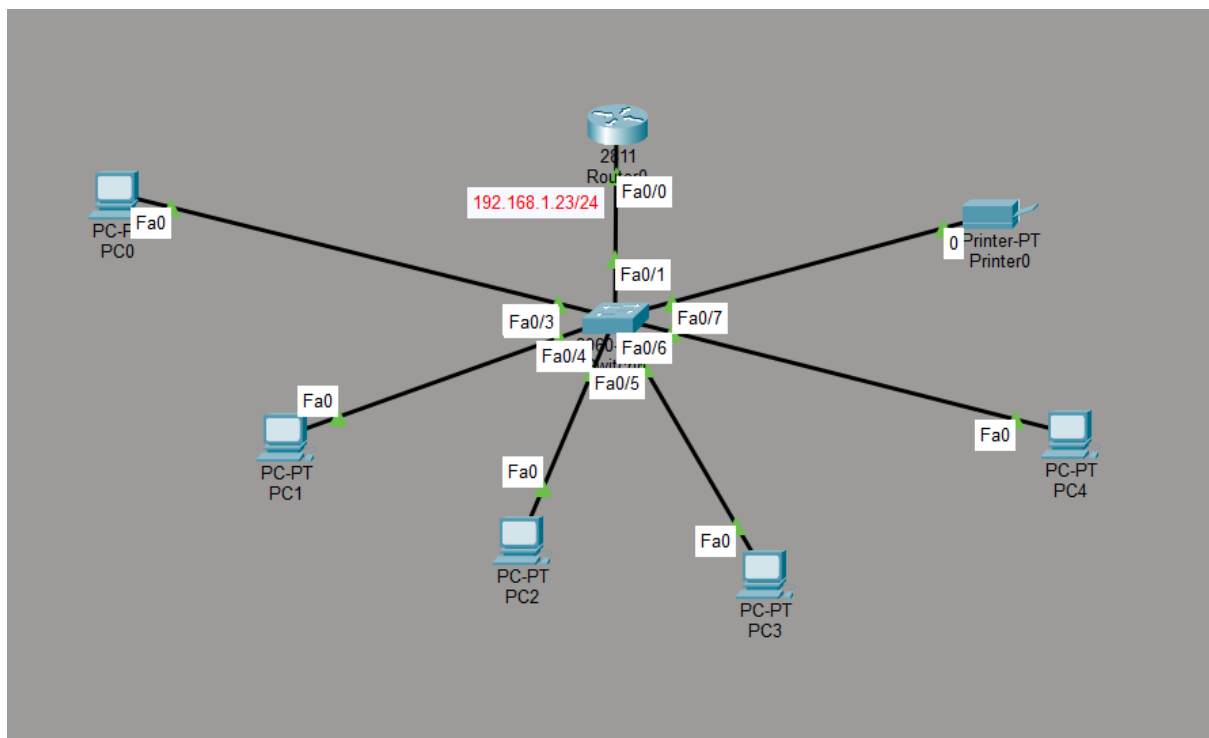
- Plug your monitor cable directly into the graphics card (not the motherboard), plug in the power cord, switch the PSU power switch to the "I" position, and press the power button on the case.

Task 2:

RAM Mismatch Troubleshooting

1. Identify the RAM Mismatch Error
2. Determine the Root Cause
3. Reseat and Correct RAM Placement
4. Reset BIOS Defaults or Adjust Profiles
5. Confirm Successful POST and Boot

Task 3:



Section C - Mini Capstone: Diagnosing Random Reboots

Three likely hardware/thermal culprits:

1. **CPU** - thermal throttling or shutdown protection triggering under sustained load in a poorly ventilated cabinet.
2. **PSU** — under heavy load (GPU/CPU draw during rendering), a marginal or aging PSU can't sustain output, causing voltage sag and reboot.
3. **RAM** — heat-induced instability (less common than CPU/PSU, but rendering workloads stress memory too, and RAM errors under thermal stress can trigger crashes/reboots).

Section D - AI Augmented Learning:

Step 1 — Prompt given to AI tool

"Generate an IP addressing and subnetting plan for a small office network with 5 systems and 1 network printer, using a Class C subnet.

AI's Output

- Network: 192.168.1.0
- Subnet mask: 255.255.255.0 (/24)
- Usable host range: 192.168.1.1 - 192.168.1.254
- Broadcast address: 192.168.1.255
- Assigned: PC1–PC5 → .1–.5, Printer → .6
- "Reserved for future expansion": .7–.254

Step 2 — Corrections I'd make working without AI:

A full /24 for 6 devices is massively oversized and wastes address space — a proper

subnetting exercise should right-size the subnet:

- Corrected subnet: 192.168.1.0/29 (255.255.255.248)
- Usable host range: 192.168.1.1 – 192.168.1.6 (6 usable hosts
- exactly fits 5 PCs + 1 printer)
- Network address: 192.168.1.0, Broadcast address:
- 192.168.1.7
- Assignment: PC1–PC5 → .1–.5, Printer → .6
- For future expansion, reserve the next block, 192.168.1.8/29
- (.9–.14), rather than leaving 248 addresses unused in the
- same broadcast domain.